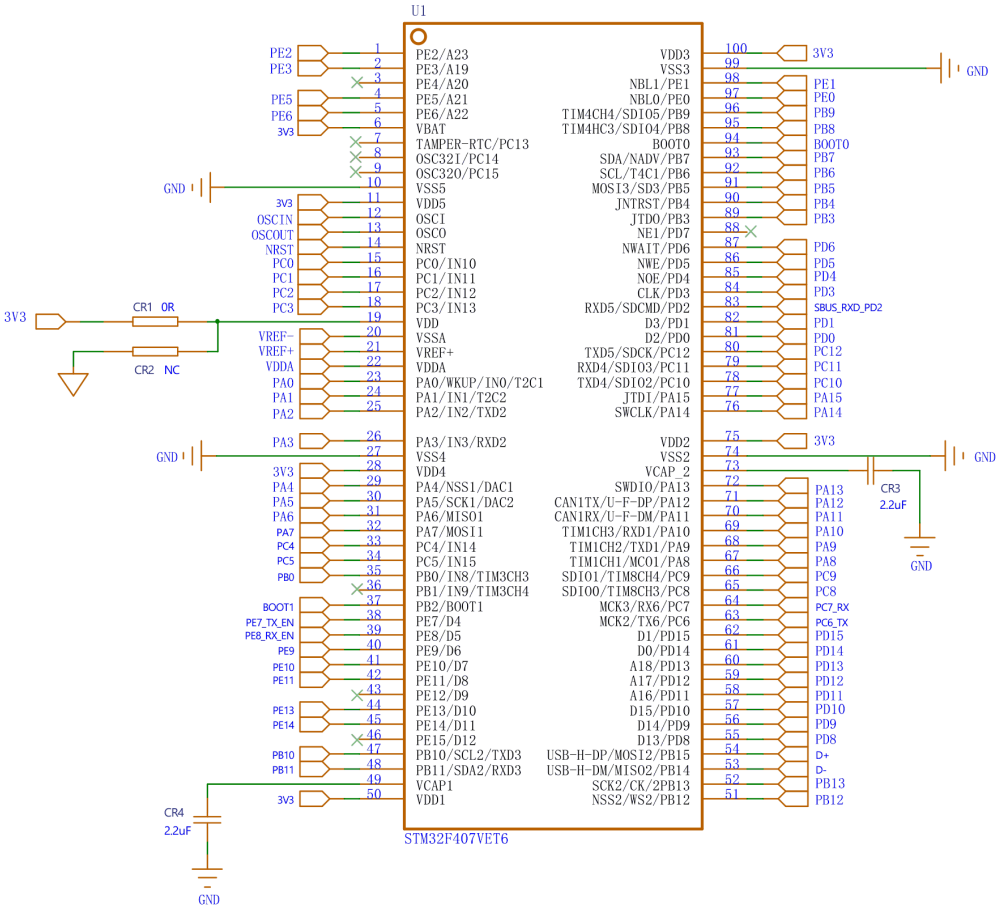
ROS Robot Controller Schematic Explanation

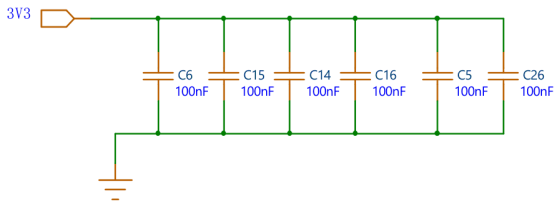
1. Main Chip Description

Main chip: STM32F407VET6



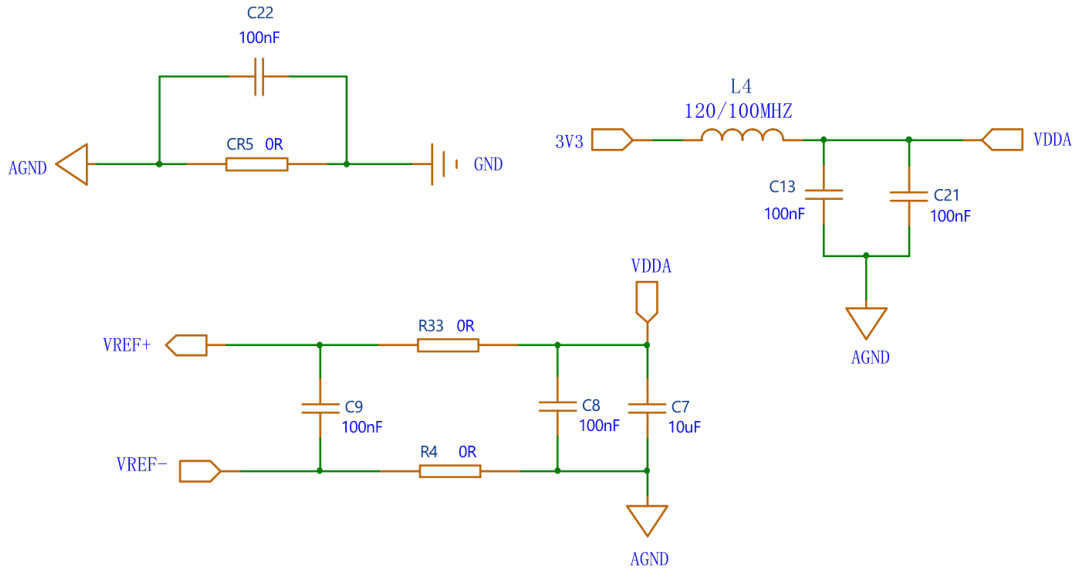
This STM32F407VET6 has 100 pins. As you can see, many of these pins are already in use, while others are left available for customers to use in their own development. Below, we will detail which pins are in use, their functions, and which pins are available.

2. Shut Capacity



These 100nf capacitors are connected to the 3.3V supply on the STM32F407VET6. The functions of these bypass capacitors are voltage stabilization and filtering.

3. Peripheral Circuitry of the VET6



VET6 Peripheral Circuitry

In the peripheral circuit of the device, the connections between AGND (analog ground) and GND (digital ground) and between 3V3 and VDDA play important roles. Analog and digital signals often need to coexist on the same circuit board, and the switching processes of digital circuits can introduce noise. Typically, two separate systems are used: one for the analog circuit (AGND)

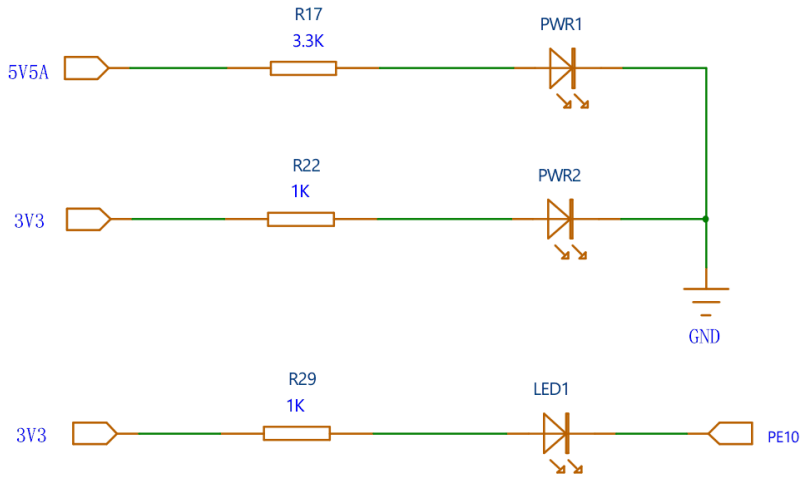
and one for the digital circuit (GND), to minimize interference between the ground currents of the analog and digital parts while ensuring consistent potential throughout the system to minimize noise impact.

3V3 is the supply voltage, and VDDA is the analog supply voltage. The circuit design may include power supply decoupling devices. An LC filter is formed by connecting an inductor (L) and two 100nF capacitors (C) in parallel. This filter reduces high-frequency noise in the supply voltage, thereby providing a more stable analog supply voltage (VDDA). This is crucial for power supply decoupling of peripheral circuits.

VREF+ and VREF- are connected in parallel with two 100nF capacitors, known as decoupling capacitors. These capacitors reduce high-frequency noise on power lines, smooth the power supply voltage, and provide a more stable voltage.

Additionally, a 10uF capacitor is connected to AGND through two 0-ohm resistors. This method of power supply filtering allows the 10uF capacitor to provide low-frequency decoupling. The 0-ohm resistors offer a low-impedance path between the power line and the ground line, helping to reduce the impact of power supply noise on the voltage. This improves the accuracy and performance of ADC and DAC circuits.

4. Power Indicator



Power Indicator

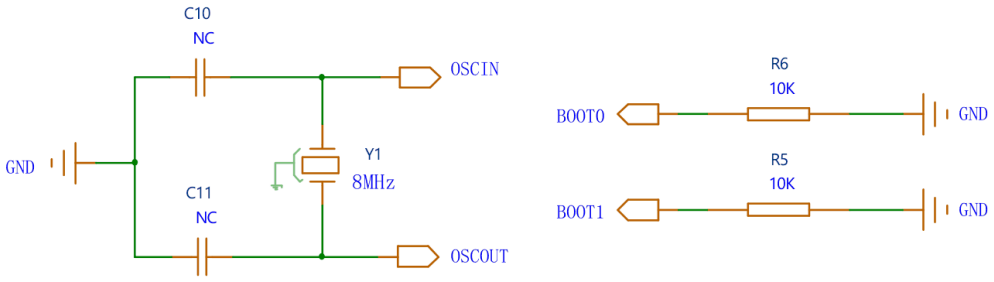
The three circuits of the power indicator light utilize resistors to regulate the current flowing through the LED, thus preventing overheating and damage to the LED.

In the first circuit, powered by a 5V supply, a 3.3k resistor is connected in series with the LED to ground. By applying Ohm's law (I = V/R), the current (I) flowing through the circuit can be calculated as I = 5V / 3.3kΩ = 1.52mA. This current value falls within the safe operating range, as the maximum continuous operating current for most LEDs is approximately 20mA.

In the second circuit, powered by a 3.3V supply, a 1k resistor is used in series with the LED to ground. Similarly, the current can be calculated as I = 3.3V / 1kΩ = 3.3mA, which is also within the safe range.

In the third circuit, this connection method is typically employed to indicate a specific status of the main chip. When the PE10 pin is configured as a low output, current flows through the LED and resistor to ground (GND), illuminating the LED. Conversely, when the PE10 pin is configured as a high-level (3.3V) output, the voltage difference across the LED is 0, causing the current to cease flowing and the LED to turn off.

5. Crystal Oscillator Circuit



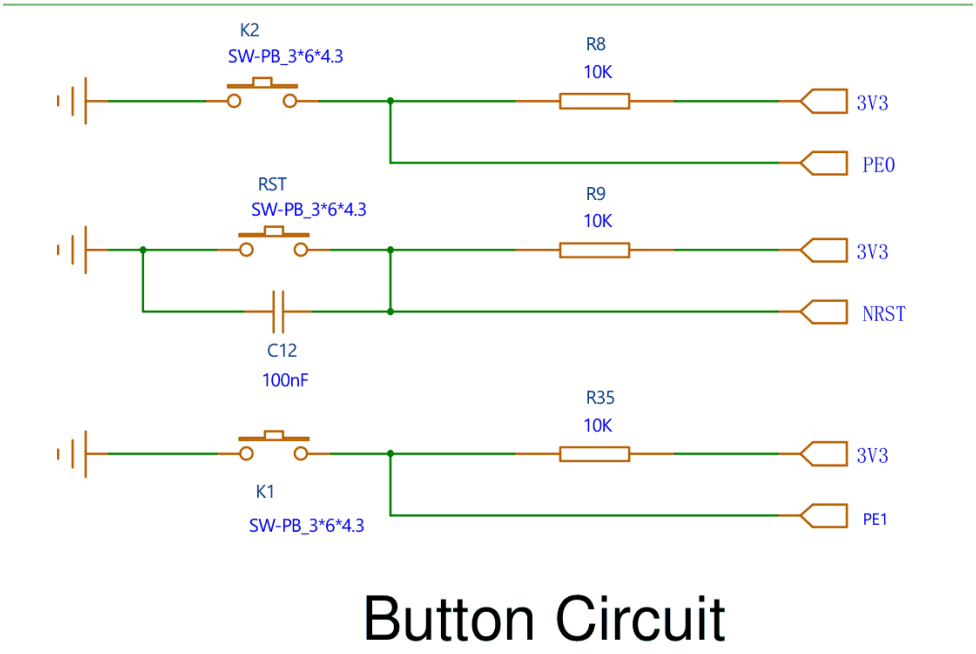
Crystal Oscillator Circuit

An 8MHz crystal is paralleled with the OSCIN and OSCOUT pins on the STM32F407VET6 chip. Each crystal oscillator pin is accompanied by a 22pF capacitor, connected to the ground wire, forming a loop crucial for initiating crystal oscillation, ensuring a stable clock signal.

Regarding BOOT and BOOT1, they serve as digital input pins to determine startup configurations upon power-on or reset. By reading the levels (high or low) at these pins, different boot sources are selected accordingly. Options include booting from built-in Flash memory, system memory, or an external memory device. Both BOOT0 and BOOT1 pins are linked to ground via 10KΩ resistors, ensuring they default to a low level during power-up or reset, typically establishing the default startup mode.

For example, the system can boot from built-in Flash memory, system memory, or an external memory device. The BOOT0 and BOOT1 pins are grounded through a 10KΩ resistor each. This means that upon power-up or reset, these pins will be pulled low (set to a low level). The purpose of this configuration is typically to set the default boot mode.

6. Button Circuit

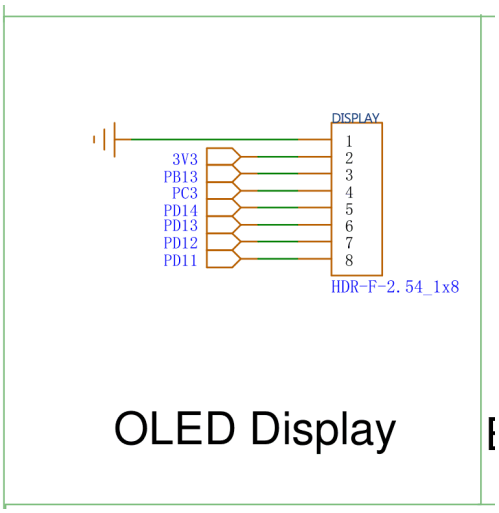


Button Circuit

These three circuits are pull-down resistor button circuits. When the button is not pressed, the PE0 pin is connected to the 3.3V power supply through a 10kΩ resistor, so the PE0 pin reads a high level. When the button is pressed, the PE0 pin is directly grounded, so the PE0 pin reads a low level.

The NRST reset button and the PE1 circuit work the same way.

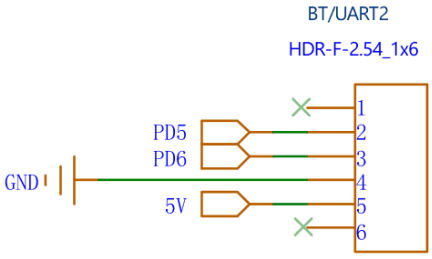
7. OLED Display



OLED Display

This socket connects to the OLED display, interfacing with the PB13, PC3, PD14, PD13, PD12, and PD11 pins of the STM32F407VET6. You can view its configuration through the program and refer to the STM32F407VET6 specification sheet in the folder to see the functions shared by these pins.

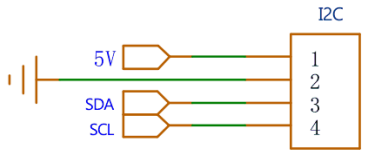
8. Bluetooth Module Port



Bluetooth Module Port

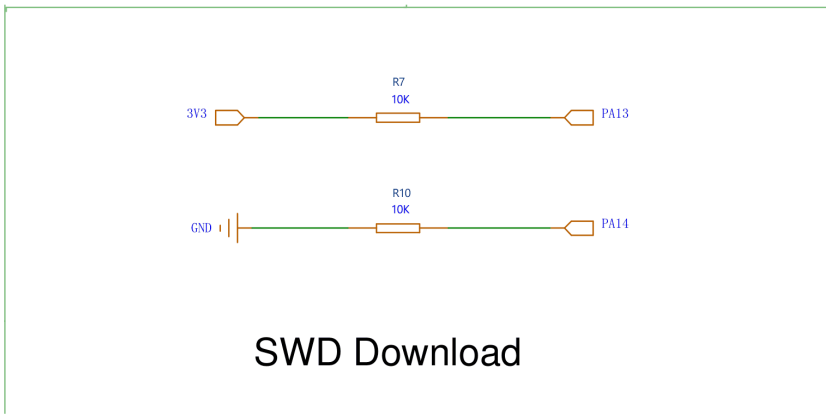
This is a screen holder connected to the Bluetooth module interface. It connects to the PD5 and PD6 pins of the STM32F407VET6. You can view the configuration through the program and refer to the STM32F407VET6 specification folder to see the functions these pins perform.

9. IIC Reserved Port



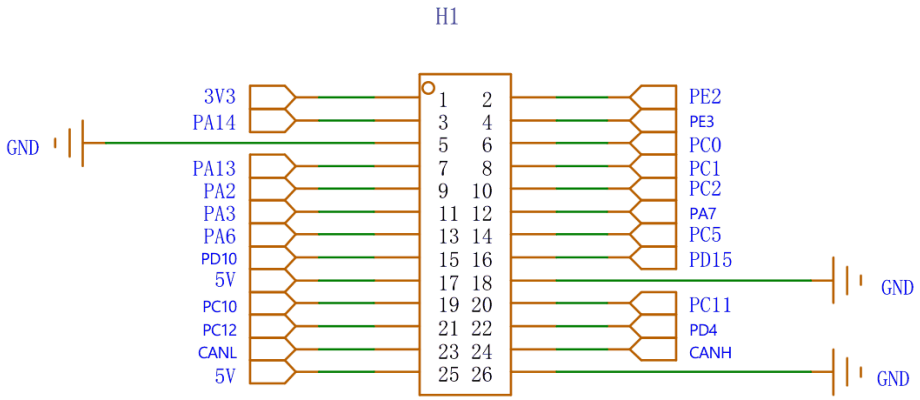
This is an IIC reserved interface that can be connected to external IIC devices, such as language modules and OLED screens. The IIC pins of the STM32 are connected here, and the IIC bus is pulled up on the MPU6050.

10. SWD Download



In the STM32F407VET6 chip, the two pins are typically used as a debug port (SWD interface), with PA13 serving as SWDIO (data line) and PA14 as SWCLK (clock line).

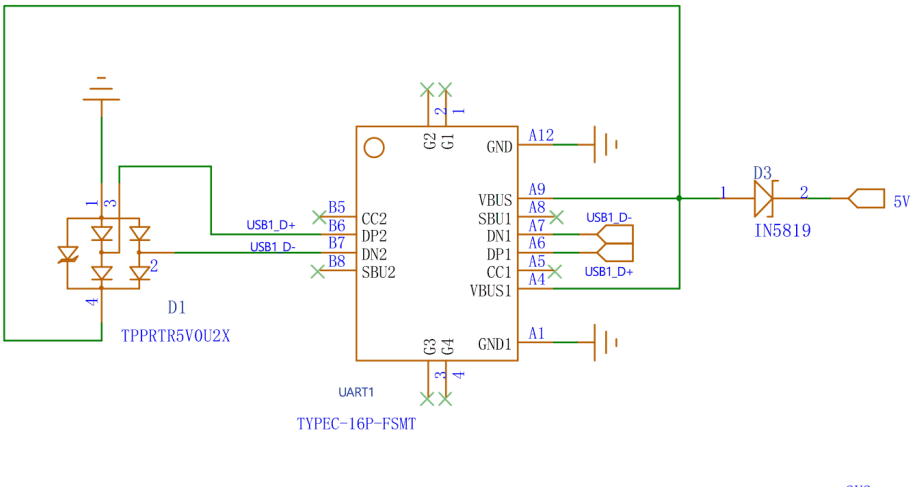
11. Reserved Port



Reserved Port

The reserved interface consists of GPIO ports on the STM32F407VET6 main chip that have not yet been used.

12. TYPEC Port



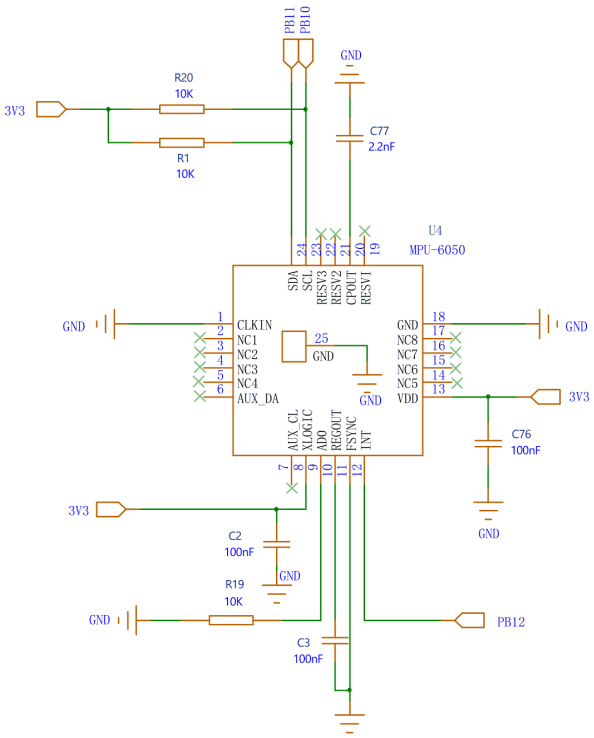
Connect a diode to VBUS to prevent reverse current on the VBUS line.

The cathode of the diode is connected to the VBUS pin, and the anode is connected to the power circuit.

This way, the power circuit can draw power from the VBUS pin but cannot supply power back to the VBUS line.

DP2 and DN2 are the differential data lines of the USB Type-C connector used for transmitting USB data. Connecting an ESD protection device can prevent these lines from being damaged by electrostatic discharge (ESD) or other overvoltage events.

13. MPU-6050



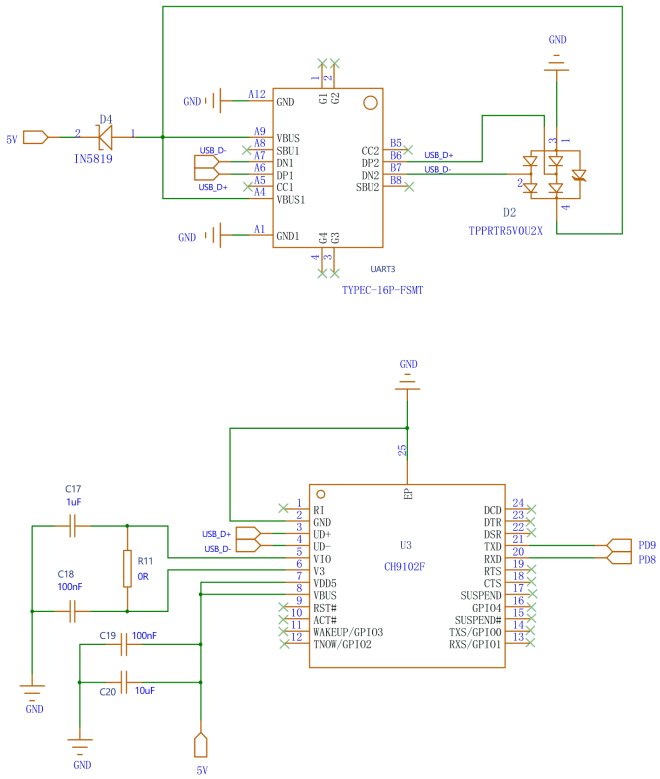
Here, both the SCL and SDA pins of the MPU-6050 are connected to 10k pull-up resistors, facilitating their functionality as pull-up resistors. Consequently, the device can pull the line low but cannot actively pull it high.
Power connection: The VCC pin of the MPU-6050 requires connection to a 3.3V power supply, while the GND pin should be linked to ground.
I2C signal line connection: SDA and SCL represent crucial lines for I2C communication, serving as data and clock lines respectively. Correspondingly, the pins associated with I2C functionality are connected to SDA and SCL on the MPU-6050. For instance, on the STM32F407VET6, PB11 can serve as

SDA, and PB10 can function as SCL.

Interrupt pin: The INT pin acts as an interrupt output. It facilitates communication between the MPU-6050 and the STM32F407VET6 chip by signaling new data availability or other relevant events.

Capacitor integration: When the MPU-6050 initiates operation and draws current from the power line, the integrated capacitor swiftly furnishes the necessary current, mitigating momentary drops in the supply voltage. Simultaneously, the capacitor serves to attenuate high-frequency noise on the power line, safeguarding the MPU-6050 against noise-induced performance disruptions.

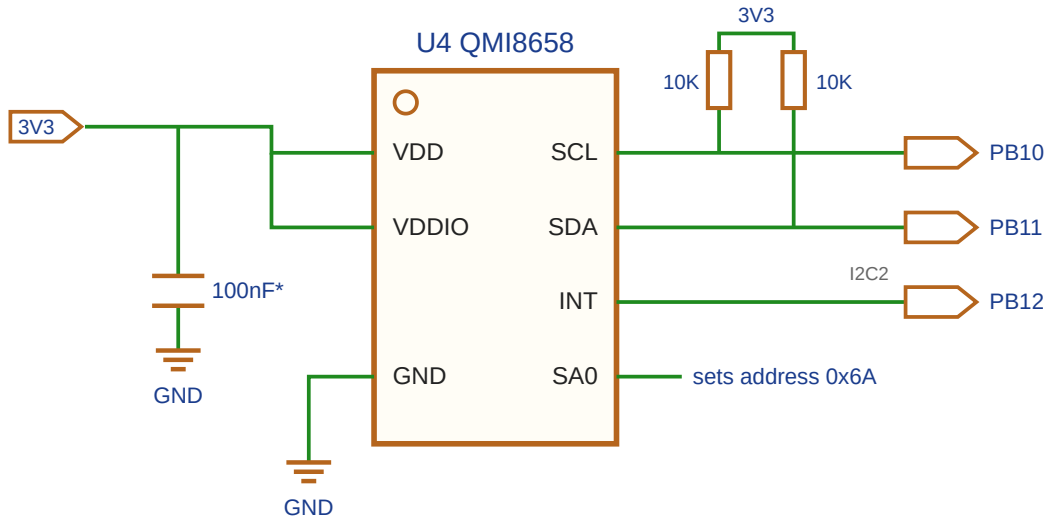
14. CH9102F Serial Port 3 Circuit



Serial Port 3 Circuit

Here, both the SCL and SDA pins of the MPU-6050 are connected to 10k

13b. QMI8658 (v1.2 boards only)



Connection drawn from Hiwonder's v1.2 example code (pins, address) and the V1.1 schematic (10K pull-ups).
Hiwonder has not published a v1.2 schematic, so this is not their drawing. *Typical decoupling capacitor, value not confirmed.

On v1.2 boards the MPU-6050 is replaced by a **QMI8658**, a 6-axis IMU made by QST: a 3-axis accelerometer and a 3-axis gyroscope, plus a temperature sensor. Like the MPU-6050, it has **no magnetometer**.

Power connection: VDD and VDDIO connect to the 3.3V supply and GND to ground. The chip works from 1.71V to 3.6V, so 3.3V is fine. A small capacitor next to the chip supplies current when it switches on and filters noise on the supply.

I2C signal line connection: the wiring is the same as the MPU-6050: **PB10 = SCL** and **PB11 = SDA**, which is the I2C2 peripheral of the STM32F407VET6. The two 10k pull-up resistors still pull the lines high; the chip can only pull them low. The 7-bit address is **0x6A** (the MPU-6050 was 0x68), set by how the SA0 pin is wired. Use standard (100 kHz) or fast (400 kHz) I2C.

Interrupt pin: the QMI8658 interrupt output goes to **PB12**, the same pin the MPU-6050 used. It can signal "new data ready", so the STM32 reads the sensor exactly when a sample is ready.

13b. QMI8658 (continued): what changes in your code

	MPU-6050 (V1.1)	**QMI8658 (v1.2)**
I2C address (7-bit)	0x68	**0x6A**
"Who am I" register	0x75, reads 0x68	**0x00, reads 0x05**
Data registers	0x3B–0x48, high byte first	**0x35–0x40** (AX_L … GZ_H), **low byte first**
Temperature	0x41–0x42	0x33–0x34
Accel ranges	±2 / 4 / 8 / 16 g	±2 / 4 / 8 / 16 g
Gyro ranges	±250 … 2000 °/s	±16 … 2048 °/s
Main set-up registers	PWR_MGMT_1, GYRO_CONFIG, ACCEL_CONFIG	CTRL1 (auto-increment), CTRL2 (accel), CTRL3 (gyro), CTRL7 (turn sensors on)
Pins on the STM32	PB10, PB11, PB12	same

So the **wiring and the CubeMX set-up do not change** (I2C2 on PB10/PB11, PB12 as an input or EXTI). Only the driver code changes: a different address, different registers, and the bytes come low byte first.

A good first test: read register 0x00. If you get **0x05**, the I2C bus and the chip both work.

Hiwonder's own driver for this chip is in their v1.2 example project, file **Hiwonder/Peripherals/QMI8658.c**, explained in "Lesson 5 RRC QMI8658 Accelerometer and Gyroscope Source Code Analysis (For V1.2 Version Only)".

Differences between V1.1 and v1.2

Hiwonder only published the V1.1 schematic. This list compares it with Hiwonder's v1.2 example code (the CubeMX project and drivers).

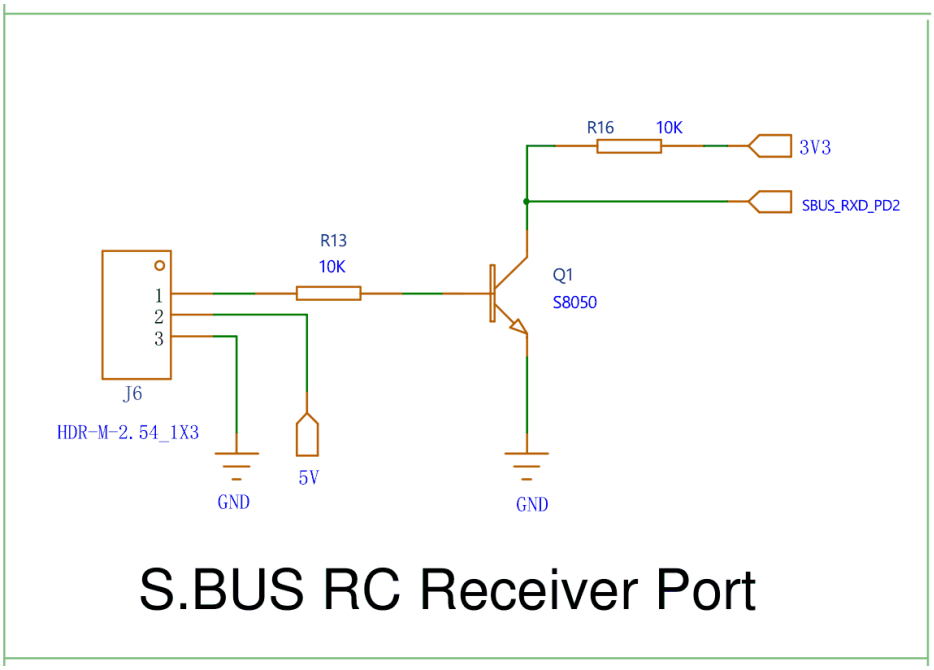
Part	V1.1 schematic	v1.2 board	How sure
IMU chip	MPU-6050	QMI8658	**Confirmed (code + "v1.2 only" lesson)**
Crystal (HSE)	8 MHz	16 MHz (PLL M = 8)	**Confirmed (code)**
IMU pins PB10 / PB11 / PB12	same	same	**Confirmed (code)**
Everything else we use: USB ports, keys, LED PE10, buzzer PA8, battery ADC PB0, encoders, servo headers, UARTs	—	same pins	**Confirmed (code matches schematic)**
Servo header voltage	all 5V	docs say 2×5V + 2×VIN	**Open: measure**
Parts that code can't show (regulators, resistor values)	—	unknown	**Can't check without a v1.2 schematic**

Note: Hiwonder's explanation text (section 17) says the buzzer is on PA4. Both the V1.1 schematic and the v1.2 code say **PA8**, so the text is out of date. Section 14 also repeats the MPU-6050 text instead of explaining the CH9102F serial chip.

pull-up resistors, facilitating their functionality as pull-up resistors. Consequently, the device can pull the line low but cannot actively pull it high.
Power connection: The VCC pin of the MPU-6050 requires connection to a 3.3V power supply, while the GND pin should be linked to ground.
I2C signal line connection: SDA and SCL represent crucial lines for I2C communication, serving as data and clock lines respectively. Correspondingly, the pins associated with I2C functionality are connected to SDA and SCL on the MPU-6050. For instance, on the STM32F407VET6, PB11 can serve as SDA, and PB10 can function as SCL.
Interrupt pin: The INT pin acts as an interrupt output. It facilitates communication between the MPU-6050 and the STM32F407VET6 chip by signaling new data availability or other relevant events.
Capacitor integration: When the MPU-6050 initiates operation and draws current from the power line, the integrated capacitor swiftly furnishes the necessary current, mitigating momentary drops in the supply voltage. Simultaneously, the capacitor serves to attenuate high-frequency noise on the power line, safeguarding the MPU-6050 against noise-induced performance disruptions.

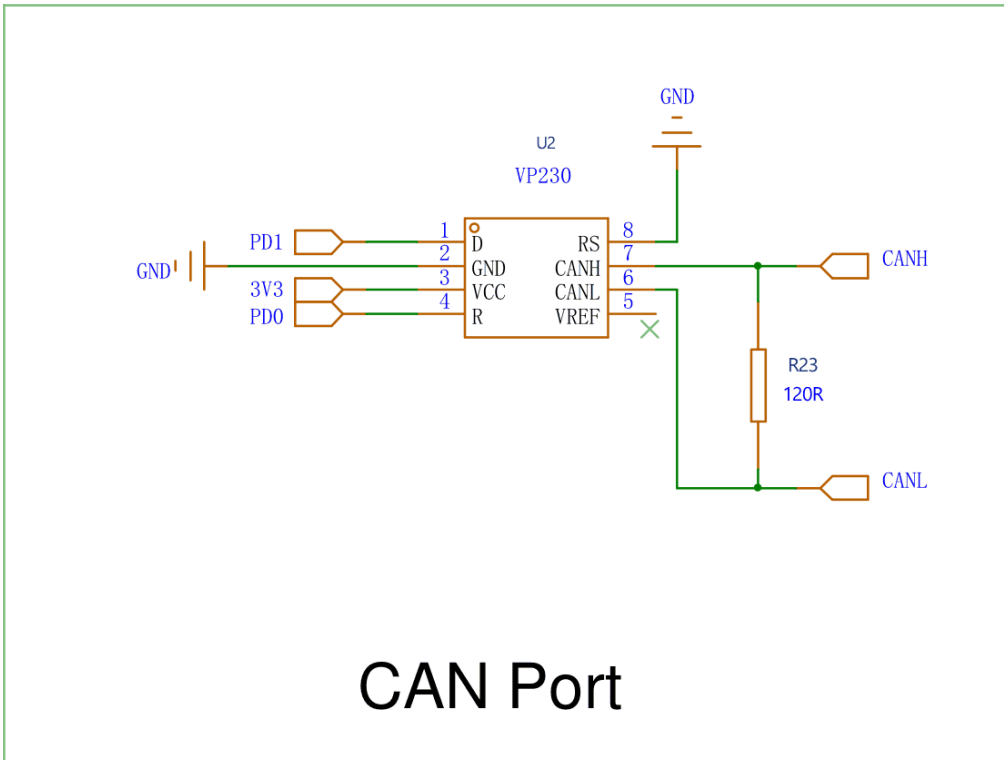
15. Aircraft Remote Control Interface for BUS Model

This serves as the remote control interface for the BUS model aircraft. An NPN transistor functions as the inverter in this setup. When the SBUS signal is at a high level, the transistor conducts, pulling the output low. Conversely, when the SBUS signal is low, the transistor switches off, and a pull-up resistor raises the output to a high level. This configuration yields the inverted SBUS signal. To receive and transmit this signal to the main chip, connect the PD2 pin of the main chip accordingly.



S.BUS RC Receiver Port

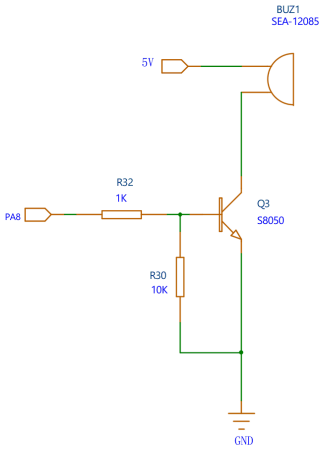
16. CAN Port



CAN Port

The CAN interface is linked to a 120-ohm resistor. This resistor serves as the terminal resistor for CAN communication, aiding in impedance matching and mitigating echo emissions.

17. Buzzer



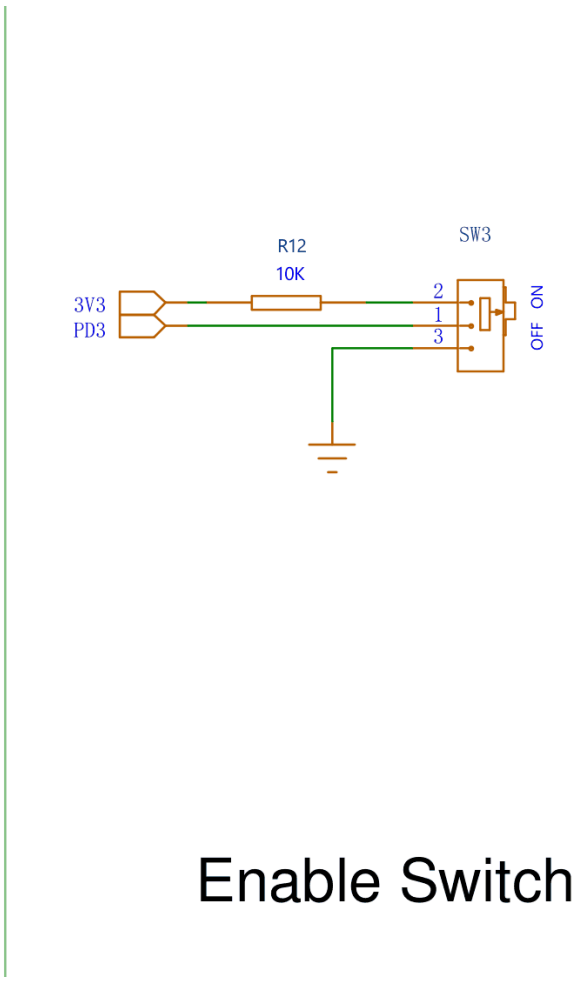
Buzzer

One end of the buzzer is connected to a 5V power source, and the other end is linked to the collector of an S8050 transistor. The collector is grounded through a 10kΩ resistor.

The base of the transistor is connected to the PA4 pin of the microcontroller via a 1kΩ resistor.

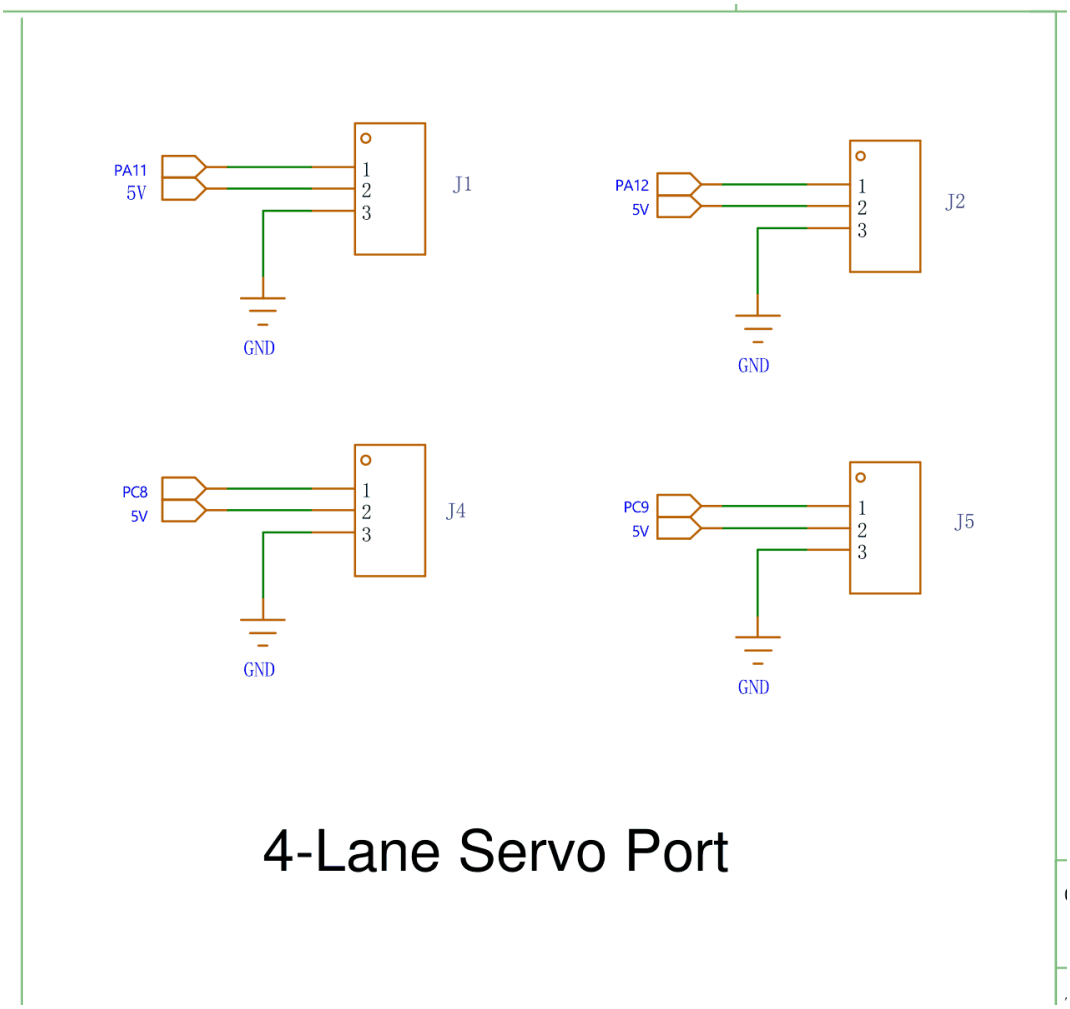
When you want the buzzer to sound, you can set the PA4 pin of the main chip to output a high level, which turns on the transistor, allowing current to flow through the buzzer and producing sound. When you want the buzzer to stop sounding, you can set the PA4 pin to output a low level, which turns off the transistor, cutting off the current flow through the buzzer and stopping the sound. This buzzer serves as an external device.

18. Enable Switch



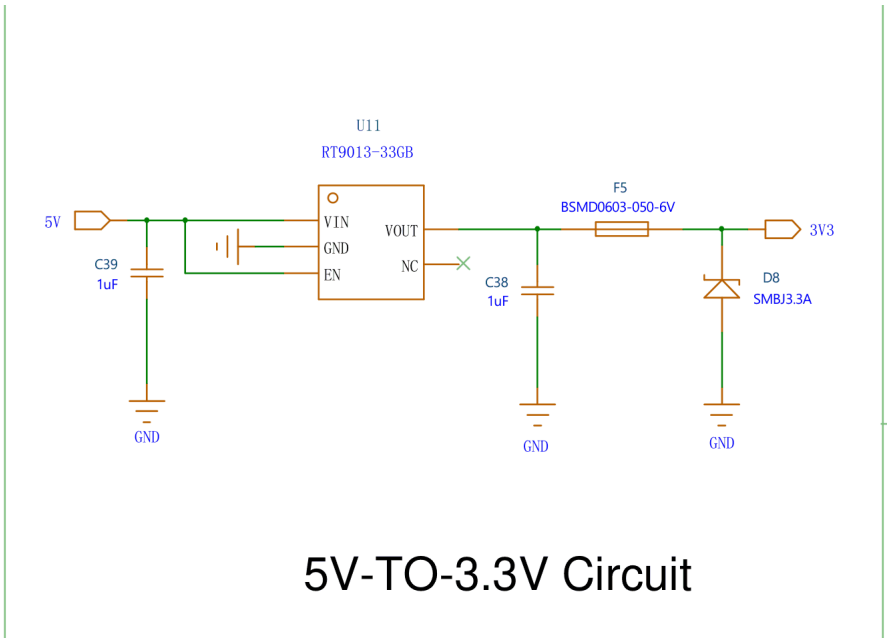
This switch is used to control the car to stop.

19. 4-Lane Servo Port



These four interfaces are designated for connecting servos, enabling servo control. There are two servo sockets connected to the 5V power source and two servo sockets connected to VIN. The 5V socket is intended for low-voltage servos, while VIN is designated for high-voltage servos powered by a high-voltage power supply.

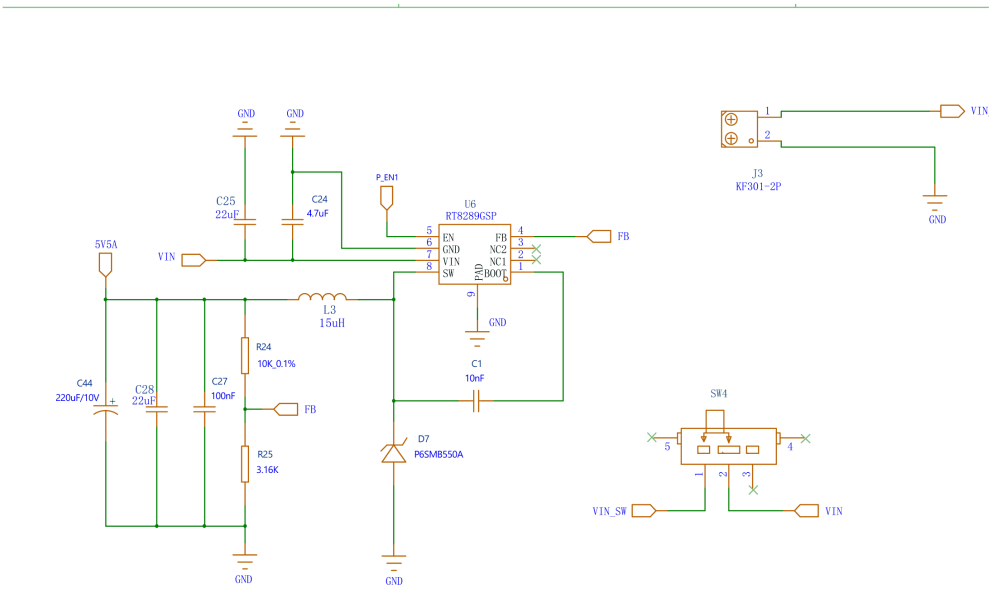
20. 5V-TO-3.3V Circuit



5V-TO-3.3V Circuit

The BSMD0603-050-6V is a fuse connected to the output terminal of the RT9013-33GB. It protects against damage from excessive current. When the current exceeds the rated value of the fuse, it will disconnect, thereby cutting off the current flow.

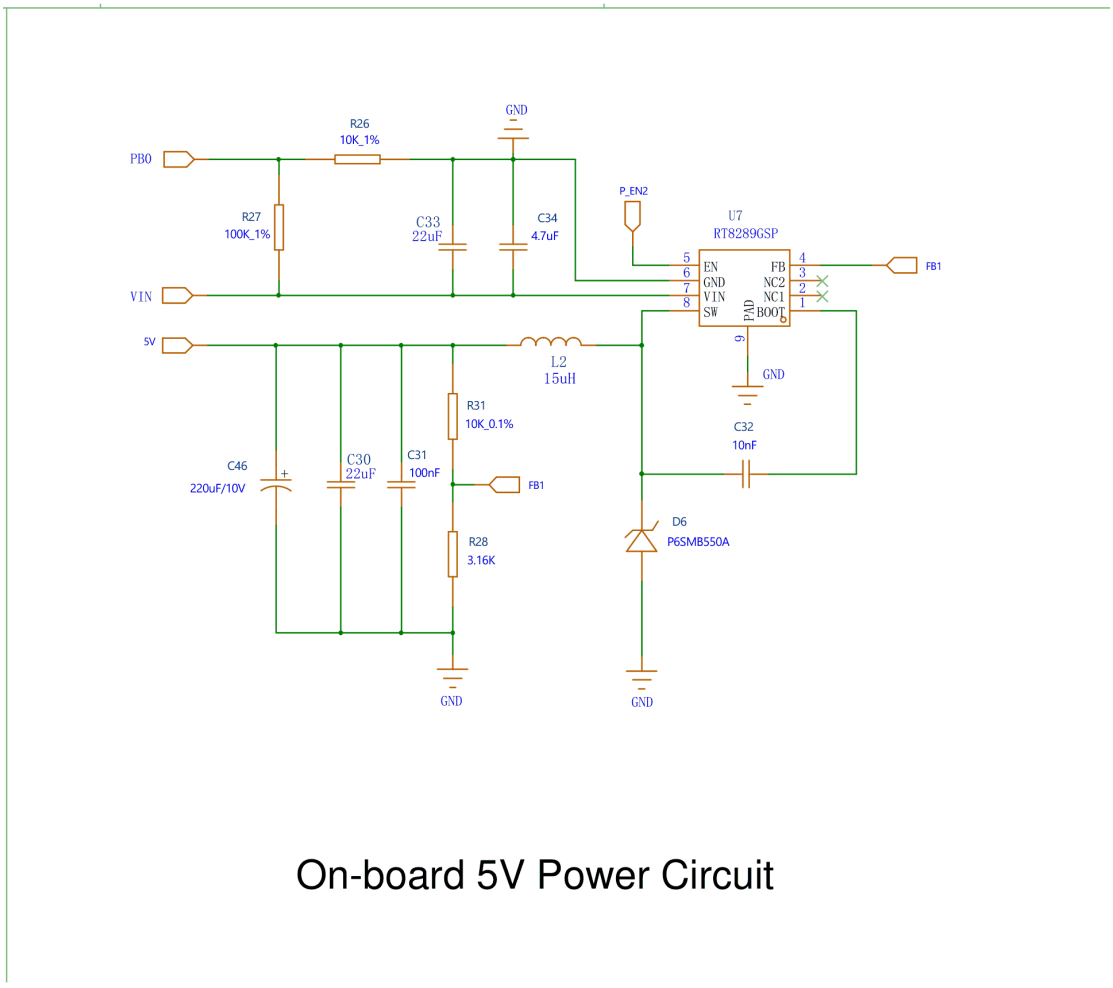
21. Raspberry Pi 5V Power Supply Circuit and Onboard 5V Power Circuit



As the name implies, this circuit supplies power to our Raspberry Pi using an

RT8389GSP power chip. The output, 5V5A, can be adjusted via R24 and R25 according to the formula: V_OUT = V_FB * (1 + R1/R2). Here, V_OUT represents the regulator's output voltage, while V_FB denotes the feedback reference voltage, typically a fixed value internally set in the regulator, often around 1.222V. (For further details, consult the RT8289GSP manual if interested.) SW4 serves as the control switch. Simultaneously, this circuit functions as an onboard 5V power supply, adjustable similarly to the aforementioned setup.

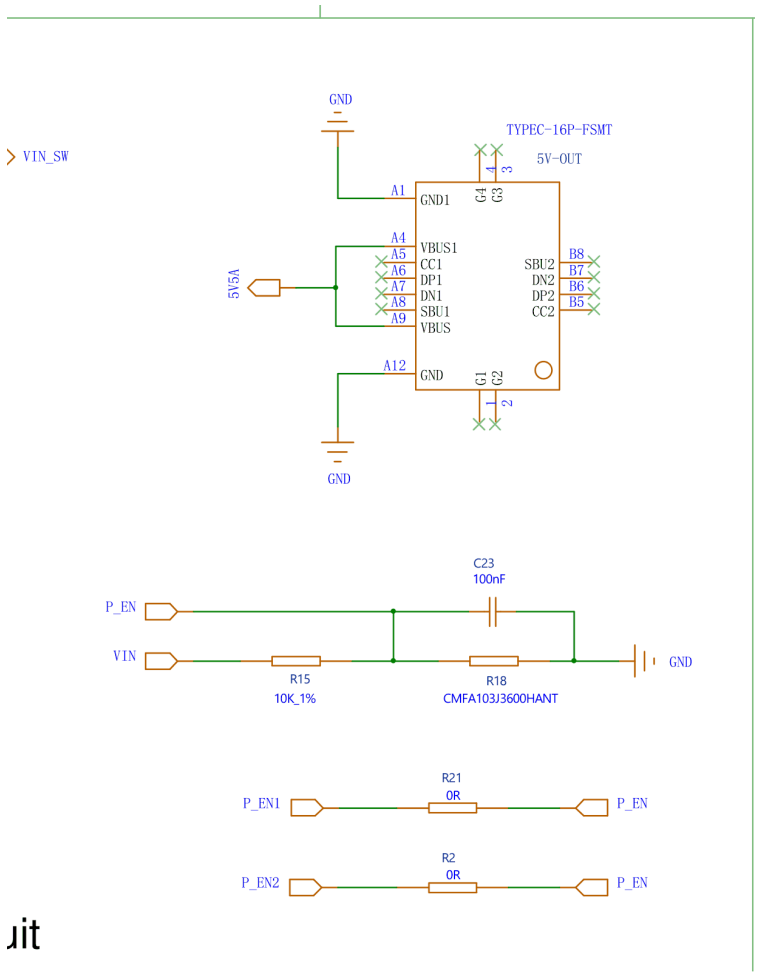
22. On-board 5V Power Circuit



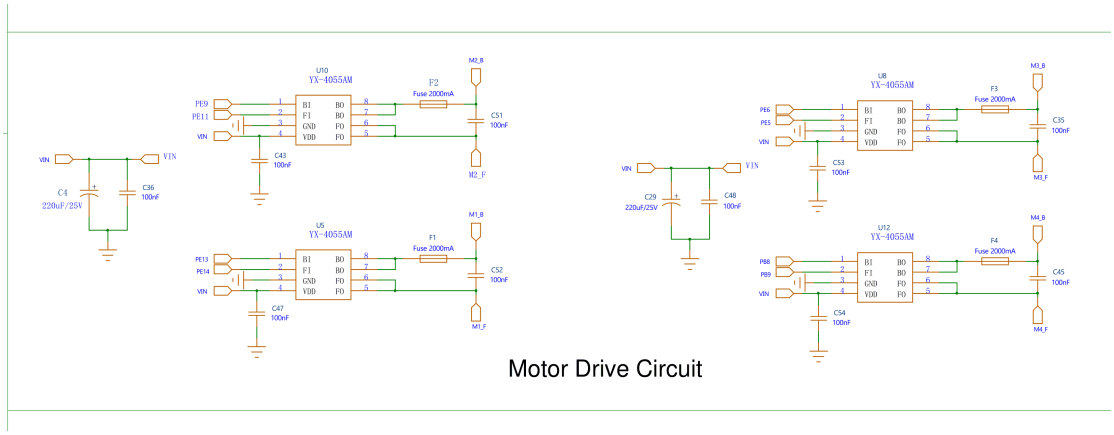
On-board 5V Power Circuit

Simultaneously, a TYEC is employed to link 5V5A to the Raspberry Pi power supply circuit. Additionally, a 10k resistor and capacitor are combined to create

a low-pass filter, aiding in the elimination of high-frequency noise. Finally, a 0-ohm jumper resistor connects both P_EN and VIN.



23. Motor Drive Circuit



Motor Drive Circuit

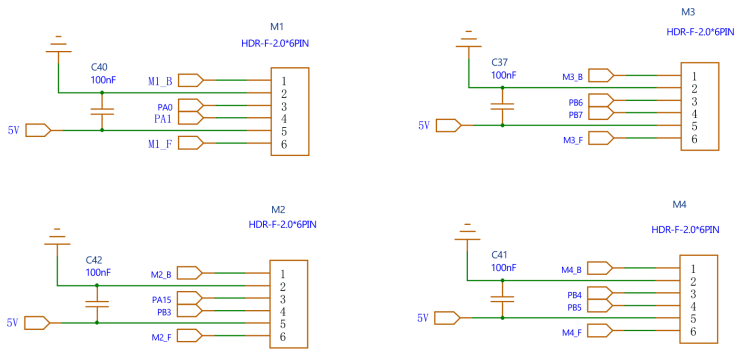
Here, we utilize the YX-4055AM motor driver chip to manage four motors. The chips, namely PA9, PA11, PE5, PE6, PE13, PE14, PB8, and PB9, connected

to the main chip, regulate our motor rotation, speed, and other functions. Furthermore, a 1500mA fuse has been added to the BO output of the motor driver chip to prevent potential chip damage due to excessive current and voltage.

Additionally, capacitors C4, C36, C29, and C48 serve the purposes of filtering and denoising. These capacitors act as filters, aiding in noise reduction along the power line. Specifically, a 220uF capacitor effectively mitigates low-frequency noise, while a 100nF capacitor excels at removing high-frequency noise.

Moreover, these capacitors play a crucial role in stabilizing the supply voltage. Fluctuations in load current can cause voltage instability, but the capacitors step in to provide the necessary current, thus stabilizing the power supply voltage and preventing voltage transients. Additionally, they prevent the power supply peak current from reaching excessively high levels. When the circuit experiences sudden demand for a large current, the capacitors supply this additional current, preventing spikes in power supply current that could otherwise impact the system.

Lastly, the following section outlines the interface for our encoded motor, which connects to the output pin of the motor driver chip and undergoes filtering via a 100nF capacitor.



Motor Encoder Circuit

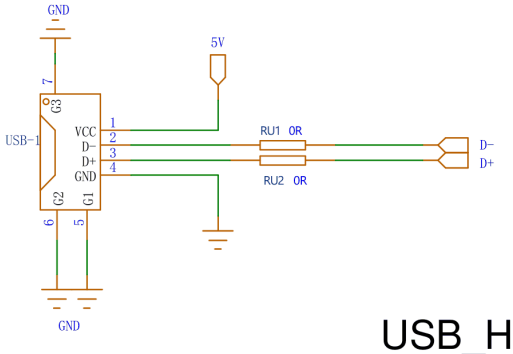
24. Serial Bus Servo Port



Serial Bus Servo Port

This circuit and interface correspond to our bus servo. PE7, PG6_TX, and PC7_RX are connected in series to the main chip.

25. USB_HOST Port Circuit



USB_HOST Port Circuit

D- and D+ are connected to the main chip via PB15 and PB14, with a jumper resistor in between. These connections receive the signal from the controller and then transmit it to the STM32 chip for decoding and processing.